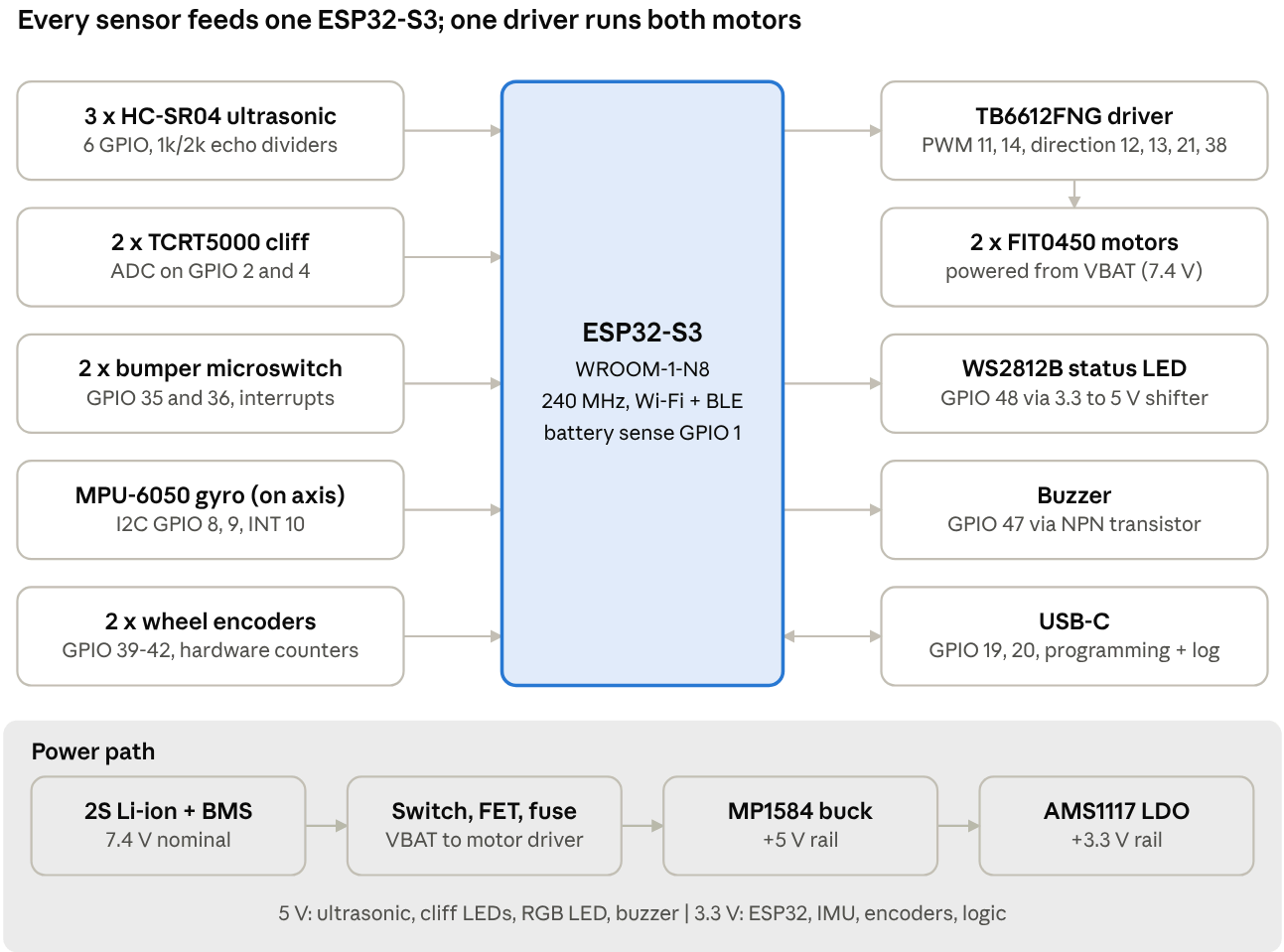
Obstacle Avoider – Architecture, Components & Algorithm

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One ESP32-S3 fuses three ultrasonic sensors, two cliff sensors, a bumper, a gyro and wheel encoders, and drives a round, centre-axle robot through a five-state machine that stops at 25 cm, turns by measured angle, and resumes only above 40 cm.

1. System architecture

The robot is built as **five subsystems around one controller**: sensing, processing, motion, user interface and power. Every sensor reports to the ESP32-S3, and the TB6612FNG is the only path to the motors.



system architecture · inputs, controller, outputs, power path

Subsystems

Subsystem	Parts	Job
Sensing – long range	3 × HC-SR04 at 0°, +40°, −40°, tilted 8° up	Distance to obstacles left, front, right (2–400 cm)
Sensing – safety	2 × TCRT5000 cliff sensors, 2 × bumper microswitches	Floor edges; contact with anything the ultrasonics miss
Sensing – motion	MPU-6050 gyro on the turning axis, 2 × hall encoders	Real turn angle; wheel speed and distance
Processing	ESP32-S3-WROOM-1	Reads all inputs, runs the state machine, drives outputs
Motion	TB6612FNG + 2 × FIT0450 encoder gear motors, differential drive	Forward, reverse, spin in place
User interface	WS2812B LED, buzzer, BOOT/START + RESET buttons, USB-C	Shows state, alerts, programming and live log
Power	2S 18650 + BMS → switch → reverse-polarity FET → 3 A fuse → VBAT; MP1584 → 5 V; AMS1117 → 3.3 V	Three rails: VBAT for motors, 5 V and 3.3 V for electronics

Mechanical architecture

- **Round 190 mm body, axle through the centre.** The bot spins in place inside its own outline (turning sweep 105 mm including the bumper).
- **Two decks.** The motors and battery sit under the base plate; the PCB sits on standoffs centred on the turning axis, under a vented top deck.
- **Three ground contacts.** Two driven wheels plus a front caster; a rear skid 3 mm above the floor stops it tipping back.

One control loop (every 50 ms)

1. Fire the three ultrasonic sensors one after another and take the median of three echoes each.
2. Read the cliff ADCs, bumper switches, gyro rate, encoder counts and battery voltage.

3. Check the safety inputs first (bumper, cliff), then run the state machine on the three distances.
4. Set the two motor PWM duties and directions; correct the speeds from the encoders.
5. Update the LED colour and buzzer, and print one log line over USB.

2. Component choices

Each choice was made against the alternatives below; the deciding reason is in the last column. Values are typical datasheet figures.

Decision	Chosen	Alternatives considered	Why the chosen part wins
Microcontroller	**ESP32-S3-WROOM-1-N8**	ATmega328P (Arduino Uno); STM32; Raspberry Pi	The design needs about 30 I/O: 6 ultrasonic, 4 encoder, 2 ADC cliff, 2 bumper, I2C, 7 motor-driver lines, LED, buzzer, USB. The ATmega328P has 20. The ESP32-S3 has hardware pulse counters for encoders, native USB, Bluetooth for telemetry, and runs in the Wokwi simulator. A Raspberry Pi needs an operating system to boot and is not real-time.

Decision	Chosen	Alternatives considered	Why the chosen part wins
Obstacle sensing	**3 × HC-SR04 fixed at 0°, ±40°**	1 × HC-SR04 on a servo; Sharp IR (GP2Y0A21); VL53L1X ToF; LiDAR	Three fixed sensors see left, front and right continuously; a servo must stop and scan (about 1 s per look) and adds a moving part. Sharp IR reads only about 10–80 cm and depends on surface colour. ToF is kept as an upgrade on the Qwiic port. LiDAR is too costly and complex for this task.
Sensor mounting	**Radial, tilted 8° up**	Level, side by side	A level sensor 60 mm above the floor sees the floor at about 22 cm (beam taken as ±15°), inside the 25 cm stop threshold. With 8° tilt the floor enters at about 49 cm.
Collision backup	**Floating bumper + 2 microswitches**	None; IR proximity ring	Ultrasonic misses soft, thin or steeply angled objects. A bumper guarantees detection on contact and needs only two GPIO.

Decision	Chosen	Alternatives considered	Why the chosen part wins
Edge detection	**2 × TCRT5000**	Downward ultrasonic; none	Ultrasonic has a 2 cm blind zone and is too large; the TCRT5000 is small, cheap and reads 2–10 mm reliably.
Turn measurement	**MPU-6050 gyro**	Timed turns; BNO055	Timed turns change with battery voltage and floor grip. The MPU-6050 is cheap, on I2C, and available in Wokwi; the BNO055 adds on-chip fusion we do not need.
Motors	**DFRobot FIT0450 (6 V, 1:120, hall encoder)**	Plain TT motors; N20 encoder motors	Encoders give speed feedback for straight driving and measured back-off. The TT format fits standard mounts and 65 mm wheels; N20 motors are weaker for this weight.

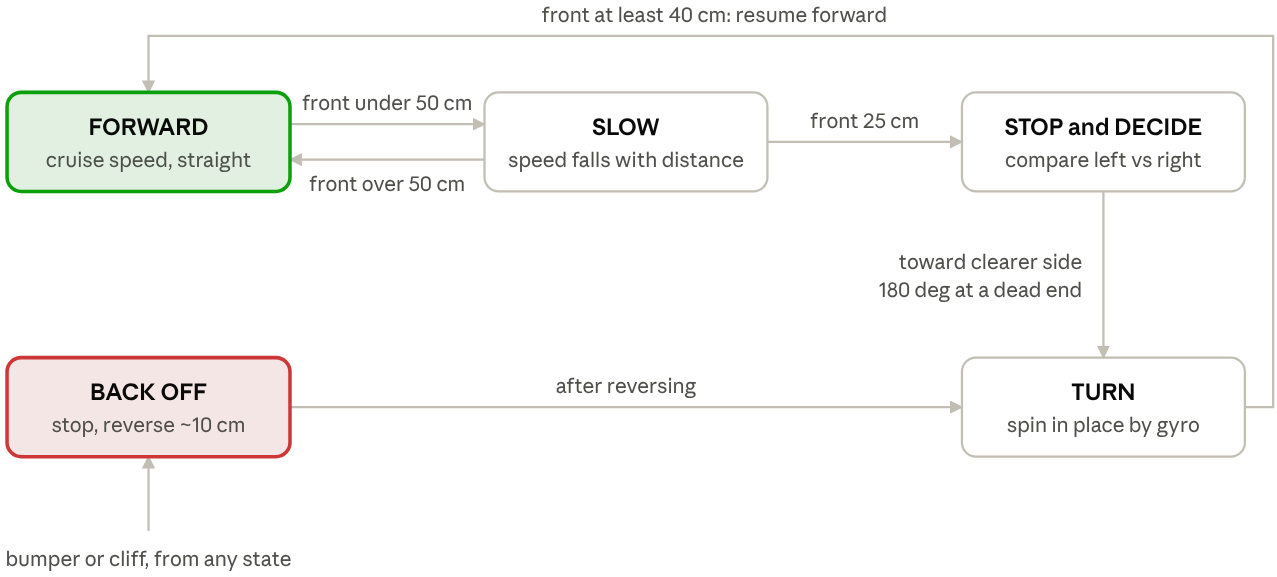
Decision	Chosen	Alternatives considered	Why the chosen part wins
Motor driver	**TB6612FNG**	L293D; L298N; DRV8833	MOSFET bridge with low voltage drop (L293D and L298N are bipolar and lose about 1.5–2 V as heat). 1.2 A per channel continuous, 3.2 A peak, logic from 3.3 V. The DRV8833 would also work; the TB6612 has a separate standby pin, which we use to keep the motors off during boot.
Drive layout	**Differential drive, 2 wheels + caster, axle at centre**	4-wheel skid steer; 3 omni wheels; car-style steering	Differential drive spins in place with two motors and simple control. Skid steer drags its wheels to turn (high current, inaccurate angles). Omni wheels need 3 motors and complex kinematics. Car steering cannot turn in place.
Chassis shape	**Round, Ø190 mm**	Rectangle, axle at rear	With the axle at the centre of a circle, nothing swings out when turning: sweep 105 mm vs 137 mm for the rectangle.

Decision	Chosen	Alternatives considered	Why the chosen part wins
Battery	**2S 18650 (7.4 V, high-drain) + BMS**	9 V battery; 4 × AA	A 9 V battery has low capacity and cannot supply motor current. AA cells sag under load. 18650 cells supply stall current; the BMS prevents over-discharge.
5 V supply	**MP1584EN buck (about 85–90 % efficient)**	7805 linear; LM2596 module	A 7805 wastes (7.4 − 5) / 7.4, about a third, of the 5 V power as heat and needs a heatsink. The MP1584 is small and needs none.
3.3 V supply	**AMS1117-3.3**	Run the ESP32 from the buck directly	A low-noise linear stage after the buck gives a clean rail for the ESP32 and IMU; 5 V in leaves enough headroom.
Protection	**P-MOSFET reverse-polarity, 3 A PTC fuse, TVS diode**	Series diode only	The MOSFET loses about 0.05 V instead of about 0.8 V for a diode; the fuse handles shorts and long stalls; the TVS clamps motor spikes.
Status output	**WS2812B LED + buzzer**	Single LEDs	One pin gives any colour per state, visible in the demo; the buzzer alerts on bump, cliff and low battery.

3. Obstacle-avoidance algorithm

The firmware is a **five-state machine with a safety layer on top**: a bumper press or cliff reading overrides every state, and the three ultrasonic distances drive the normal states.

Safety checks override everything; the bot resumes only above 40 cm



obstacle-avoidance state machine · 5 states

States

State	Entered when	What the robot does	Leaves when
FORWARD	Start, or front ≥ 40 cm after a turn	Both wheels at cruise speed; encoders keep them equal; steers away from a side wall under 20 cm	Front < 50 cm → SLOW
SLOW	Front < 50 cm	Speed drops in proportion to the front distance	Front > 50 cm → FORWARD; front ≤ 25 cm → STOP
STOP and DECIDE	Front ≤ 25 cm	Stops for 0.3 s, compares left and right distances	Always → TURN (toward the larger distance; 180° if all three < 25 cm)

State	Entered when	What the robot does	Leaves when
TURN	After STOP or BACK OFF	Spins in place; integrates gyro rate into an angle	Target angle reached and front ⩾ 40 cm → FORWARD; otherwise turns another 45°
BACK OFF	Bumper pressed or cliff seen, from any state	Stops, reverses about 10 cm measured by encoders	Distance done → TURN 90° away from the triggered side

Pseudocode

```
every 50 ms:
    L, F, R   = median_of_3(ultrasonic left, front, right)   # fired one
after another
    bump_side = read_bumpers()        # none / left / right
    cliff     = read_cliff_adc()      # none / left / right
    gyro_z    = read_gyro_rate()

    if (bump_side or cliff) and state != BACK_OFF:
        state = BACK_OFF; turn_dir = away_from(bump_side or cliff)

    switch state:
      FORWARD:  drive(CRUISE); steer_away_if(L < 20 cm or R < 20 cm)
                if F < 50 cm: state = SLOW
      SLOW:     drive(lerp(SLOW_MIN, CRUISE, (F - 25 cm) / 25 cm))
                if F >= 50 cm: state = FORWARD
                if F <= 25 cm: stop(); state = STOP
      STOP:     wait 0.3 s
                if L, F, R all < 25 cm: turn_dir = right; target = 180 deg
                else: turn_dir = (L > R) ? left : right; target = 45 deg
                angle = 0; state = TURN
      TURN:     spin(turn_dir); angle += |gyro_z| * dt
                if angle >= target:
                    if F >= 40 cm: state = FORWARD
                    else: target += 45 deg
      BACK_OFF: drive(-BACK_SPEED) until encoders show 10 cm
                then target = 90 deg; angle = 0; state = TURN
```

```
    keep_wheels_equal_with_encoders()   # PI correction while driving
straight
    set_led(state); log(L, F, R, state)
```

Thresholds and why

Setting	Value	Why this value
Slow zone	25–50 cm	Smooth approach instead of hard stops
Stop	25 cm	Room to stop from cruise speed; turning sweep is only 10.5 cm, so it can always turn away
Resume	40 cm	15 cm of hysteresis above the stop distance prevents stop-go jitter
Side steer	20 cm	Keeps it off walls when driving alongside them
Turn step	45° (180° at a dead end)	Small steps find the nearest gap; measured by gyro, not by time
Back-off	about 10 cm	Clears the bumper and the cliff sensors from the hazard before turning
Control loop	50 ms (20 Hz)	Three sensors fired in sequence fit inside one loop
Motor PWM	20 kHz, max duty about 80 %	Inaudible; limits the 6 V motors on the 7.4 V battery

Robustness details

- **No echo** within the timeout is treated as “nothing in range” (maximum distance), not as zero.
- **Median of 3** readings rejects single bad echoes; sensors are never fired together, so they cannot hear each other.
- **Hysteresis** between stop (25 cm) and resume (40 cm) stops oscillation at the boundary.
- **Gyro turns** give the same angle at any battery level or floor grip.
- **Low battery** (below 6.8 V): speed limited, LED and buzzer warn.

The same logic was run in a Fusion 360 animation and a 2D simulation: 45 s in a room with 3 boxes and a sound-absorbing cushion, **0 collisions**, all five states used, and the bumper caught the cushion the ultrasonics could not see.

4. How it fits together

Every problem we set out to solve is handled by a matching hardware choice and algorithm feature.

Problem	Hardware / mechanical answer	Algorithm answer
Obstacle ahead (brief scenario 2)	Front HC-SR04	SLOW below 50 cm, STOP at 25 cm
Choosing a way around (scenario 3)	Left and right HC-SR04 at ±40°	STOP compares L vs R; TURN toward the larger
Resuming safely (scenario 4)	Front HC-SR04	Resume only at ⩾ 40 cm (hysteresis)
Corners hitting walls while turning	Round body, axle at centre	Turn in place without moving forward
False stops from floor echoes	8° sensor tilt	—
Soft or thin obstacles	Floating bumper + microswitches	BACK OFF overrides all states
Table edges and steps	TCRT5000 cliff sensors	BACK OFF overrides all states
Inaccurate turns	MPU-6050 on the turning axis	Gyro-integrated turn angle
Drifting off a straight line	Encoder motors	Encoder speed correction
Dead end	Three sensors cover 120° +	180° turn when all three < 25 cm
Wall alongside	Side sensors	Steer away below 20 cm
Low battery	Battery-sense divider on GPIO 1	Speed limit + LED/buzzer warning